

NotGPU Attention

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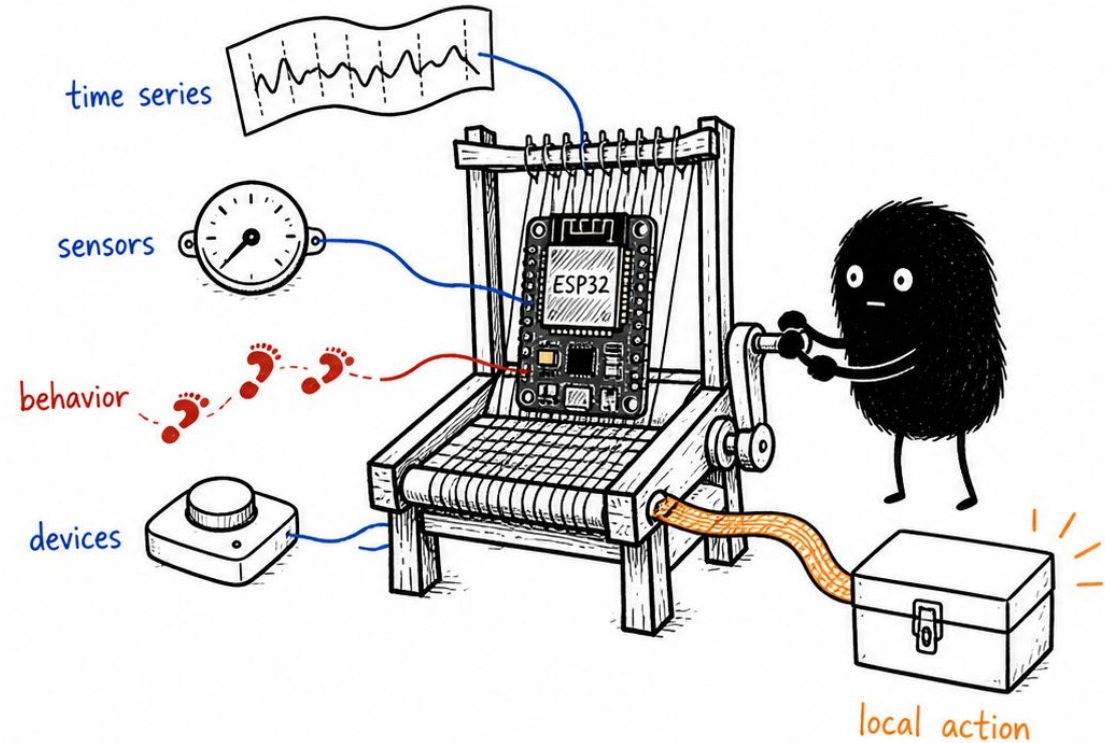
BROADER GOAL

Why we need to inference on Edge Device

Targeting edge intelligence/
tiny/IoT

- **Local evaluation on edge devices**
Smartwatches, sensors etc
- **Lower Latency**
- **For Sensitive user behavior**
Keep information at the network edge
- **Low Cost**

\$7/device



CHALLENGE STATEMENT

Make self-attention survive a microcontroller budget (ESP32-C3)

Bytes — $\log_{10}$ scale · on

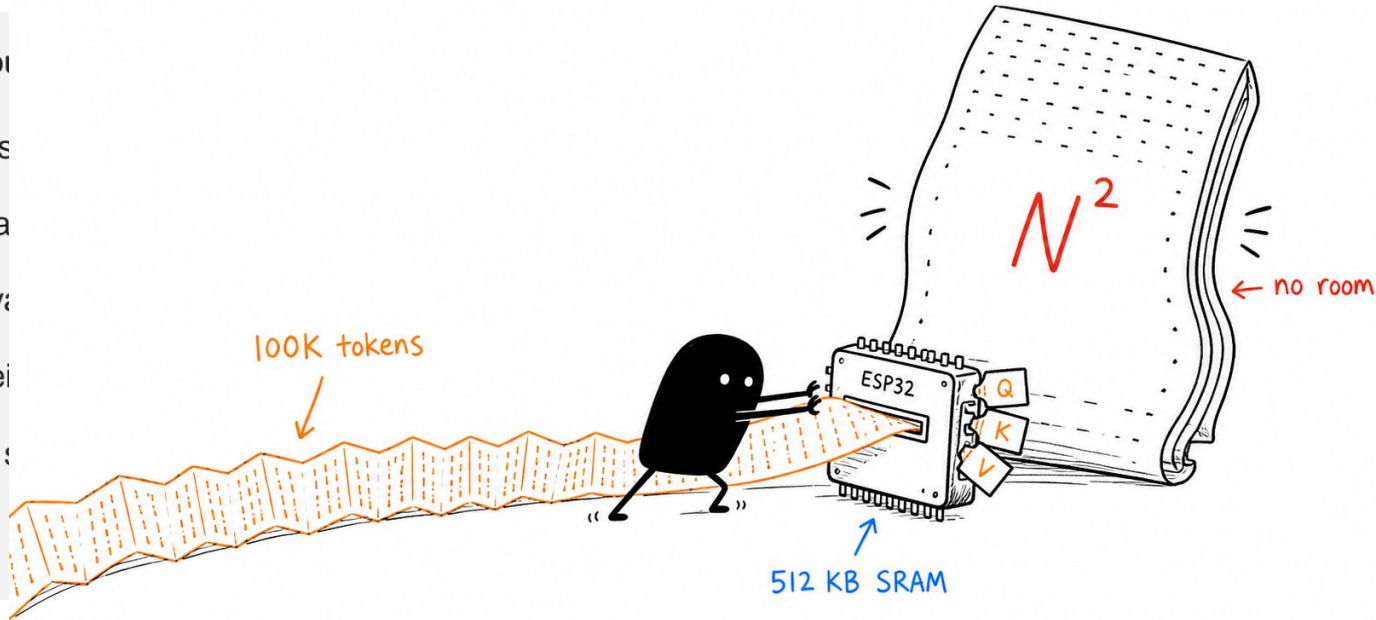
SRAM us

Case-2 a

Case 13 activa

Case 8 int16 wei

Case 14 KV s



XIAO ESP32 C3

ESP32-C3 Attention Optimization Path

One shared firmware path, progressively optimized and reused across test cases

● FINAL: OPTIMIZATION 23

FP32 / HYBRID INPUT



Shared ESP32-C3 baseline

Original forward pass

42.15 s per forward

OPTIMIZATION PATH

OPTS 1-5

Integer compute

Integer attention
Exp LUT
GEMM tiling
Integer GELU

OPTS 6-10

Quantize & fuse

QKV quantization
Integer layer norm
Output fusion

OPTS 11-19

Kernel assembly

Integer PV
core4 / core5
QK / PV assembly

OPTS 20-23

Profile & finalize

Profiling
Residual fast path
KB0 + core5 fix

OPT 23

Final shared firmware

1.996 s per forward

21.1x overall speedup



One optimized firmware path

Integer compute · fusion
quantization · assembly
profiling · fast residuals

VALIDATION ACROSS TEST CASES

OPT 23 REUSED ACROSS ALL CASES

TC1

Opt 23 reuse

21.2x single-board

TC2

Opt 23 reuse

21.1x single-board

TC3

Opt 23 reuse

21.2x single-board

TC4

Opt 23 reuse

21.2x single-board

TC5

Opt 23 reuse

21.2x single-board

TC7

Opt 23 reuse

1.03x

TC11

Opt 23 reuse

1.14x

TC12

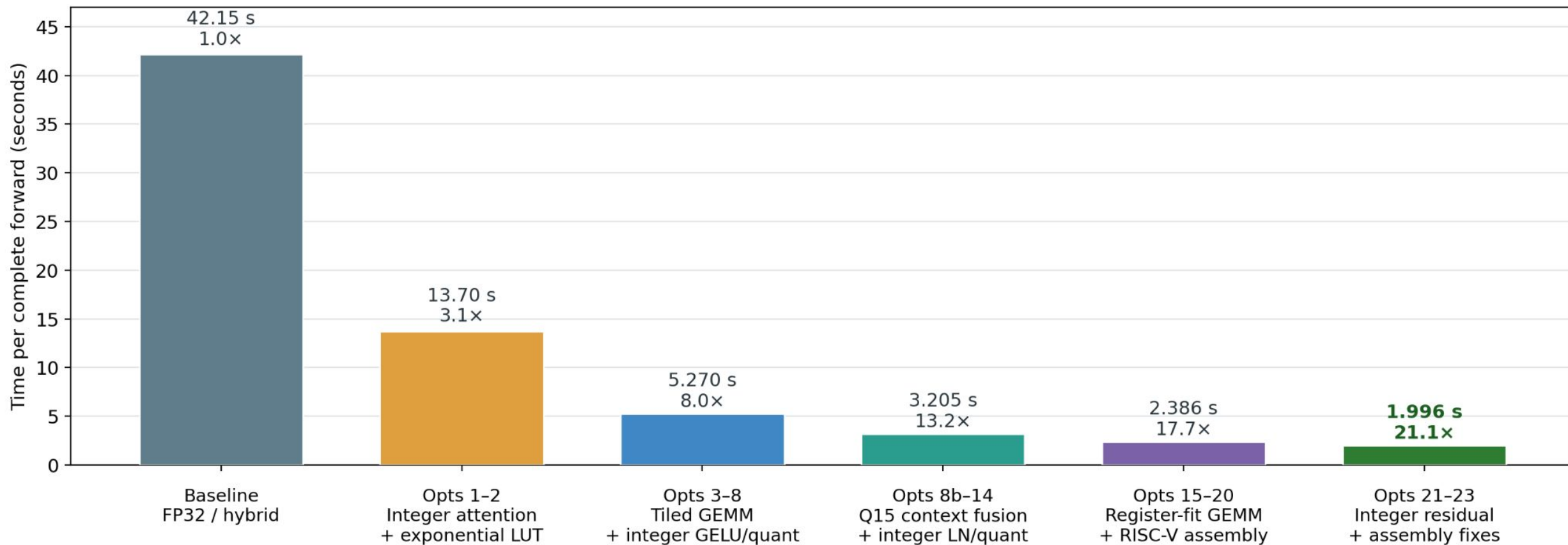
Opt 23 reuse

0.93x

PASSES

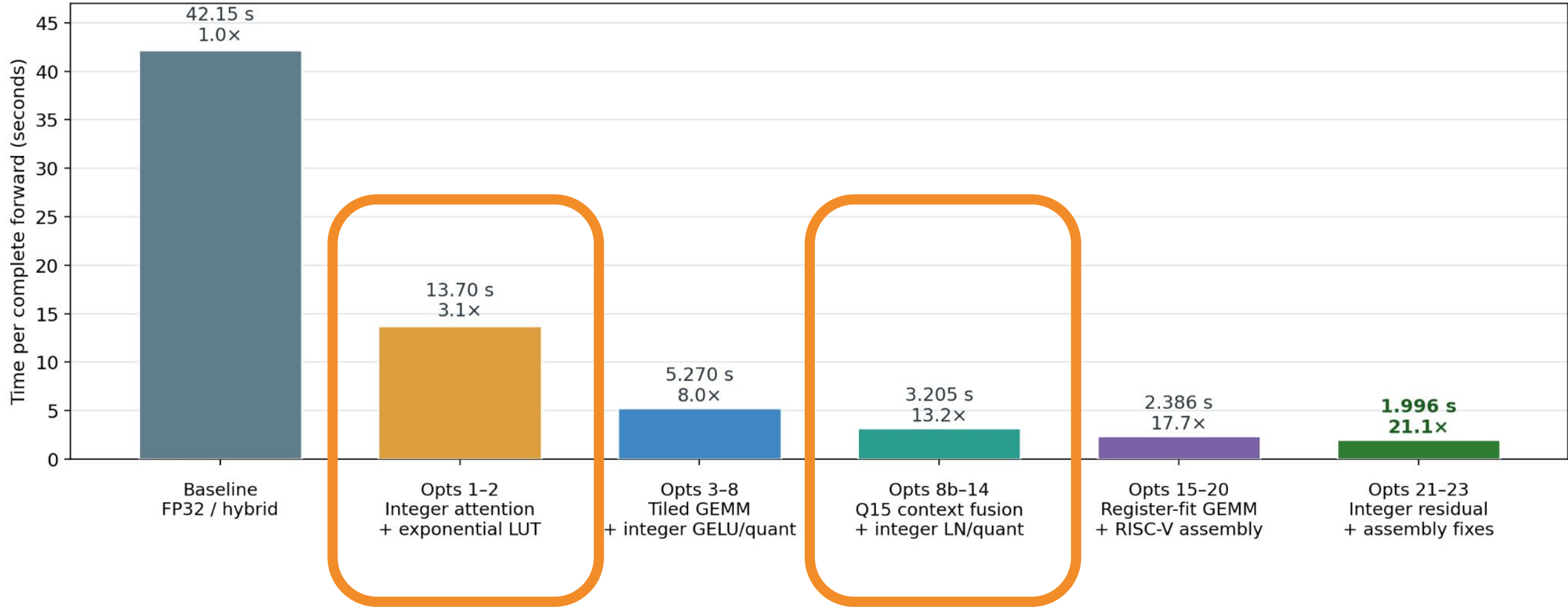
Case 2 single-board optimisation

One XIAO ESP32-C3 · B=1 · six cumulative stages · linear time scale



Case 2 single-board optimisation

One XIAO ESP32-C3 · B=1 · six cumulative stages · linear time scale



How to avoid float operation

Quantisation

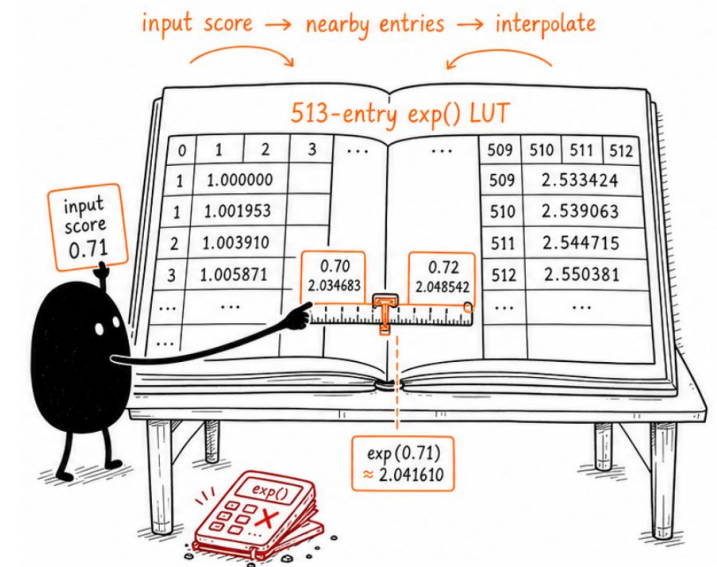
- Convert FP32 to Q15
 - Q15: 16 bit signed int

$$x_i \approx q_i \cdot \frac{\max |x|}{32767} \quad q_i \in [-32767, 32767]$$

Attention: **30s -> 1.6s (18x)**

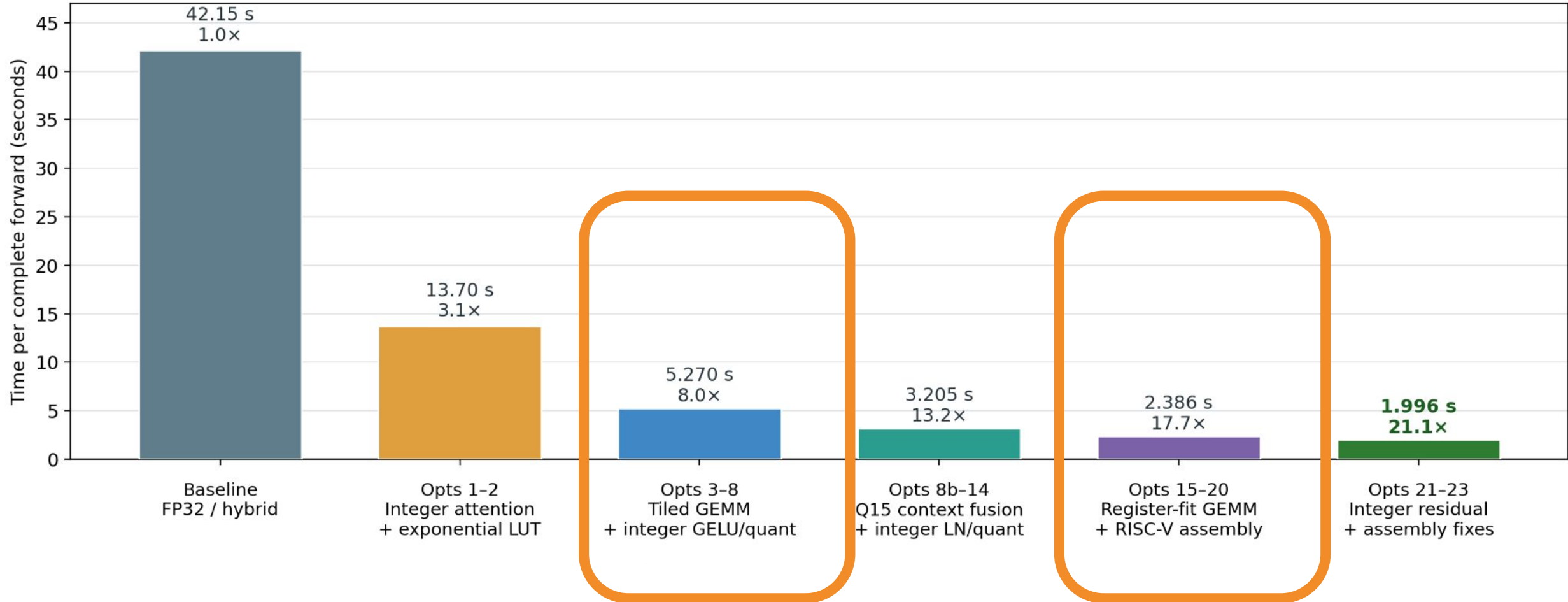
Look Up Table (LUT) for Exp

- Approximate Exponential calculation by LUT



Case 2 single-board optimisation

One XIAO ESP32-C3 · B=1 · six cumulative stages · linear time scale

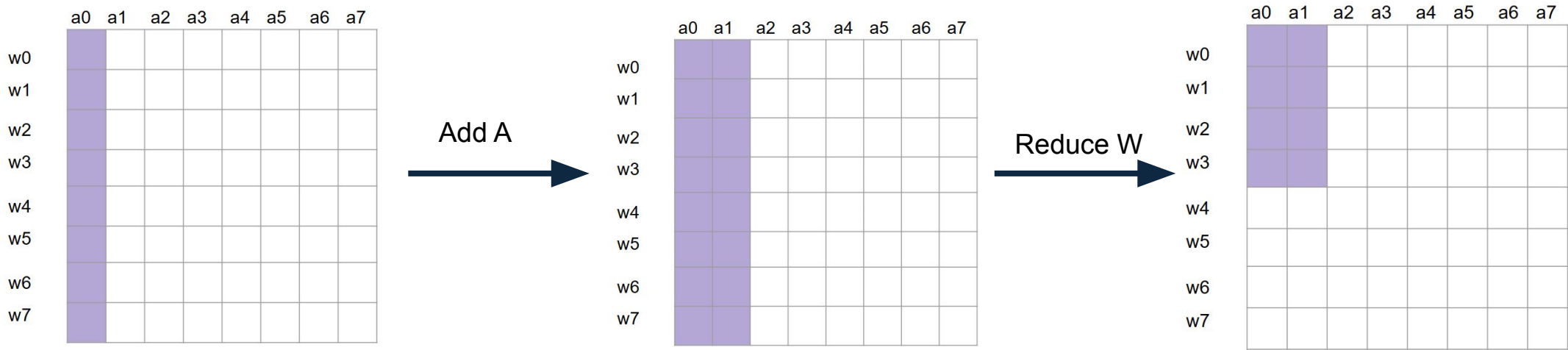


BROADER GOAL

Register optimisation for GEMM

ESP32 has **28 usable registers**
Explored the best tiling pattern

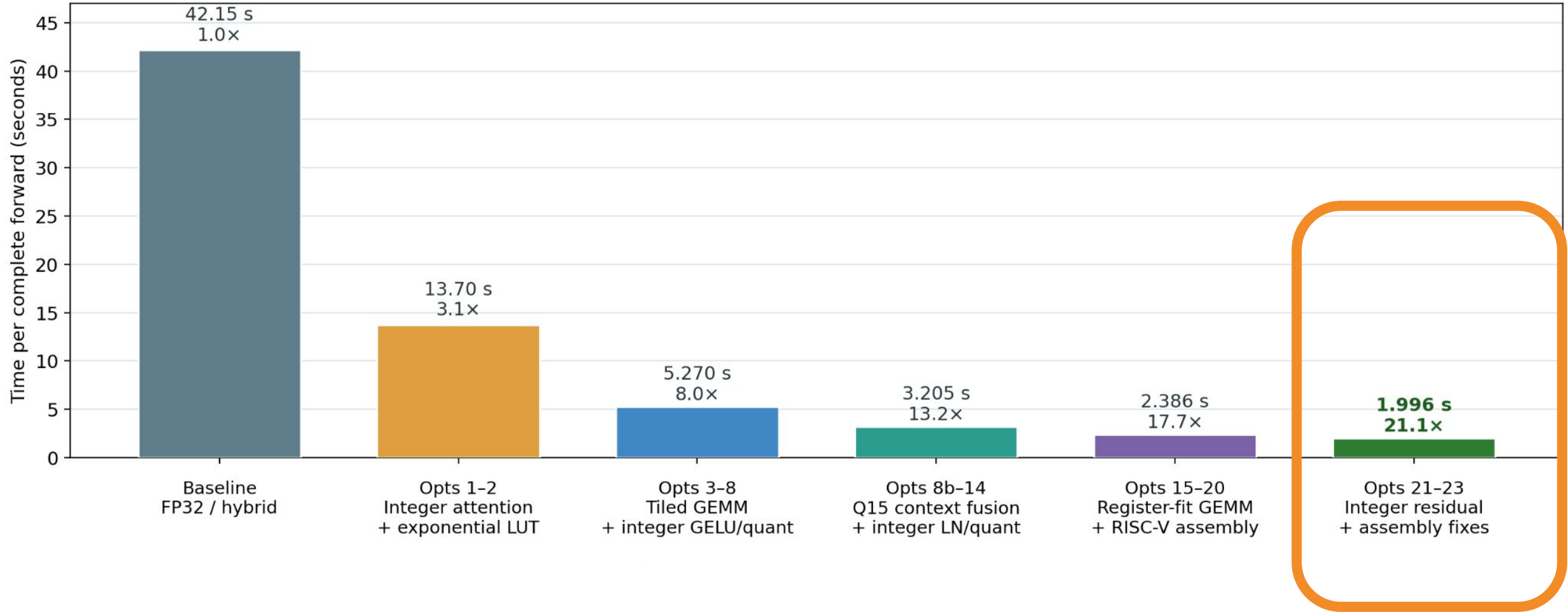
$$C_{i,j} += A_{i,k} W_{j,k}$$



	8 x 1	8 x 2	4 x 2
Output values	8	16	8
Required registers	17	36+ (spill)	20
Cycles per GEMM call	Not measured	15.69M	13.02 M
Complete Forward	Not measured	3.184 s	2.982 s

Case 2 single-board optimisation

One XIAO ESP32-C3 · B=1 · six cumulative stages · linear time scale



BROADER GOAL

Fusion & Assembly level optimisation

1 FUSION — DELETE MEMORY PASSES

Before



The same activation is materialized, read, and rewritten.

After



$xq += \text{round}((\text{acc} + \text{bias}) \times \text{scale})$

No fp32 projection buffer · res1/res2 removed · no new RAM

268 → ~10

instructions / output epilogue

86 ms → ~0

residual passes / forward

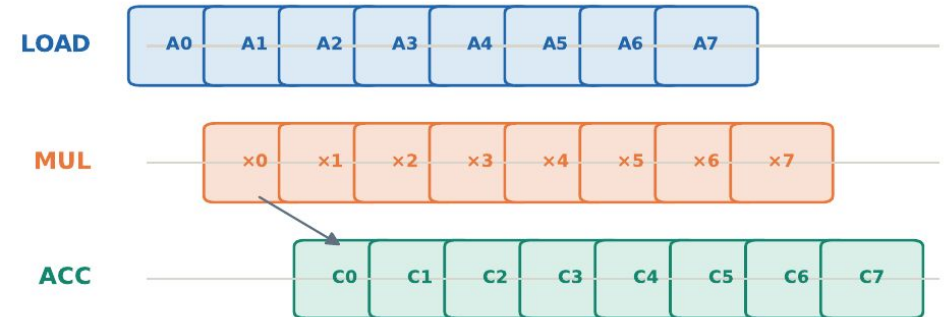
-11.0%

2.385 → 2.122 s

2 ASSEMBLY — SCHEDULE INDEPENDENT MACs

Hand-scheduled 8-row head GEMM

Load one weight, load A0...A7, then update eight accumulators.



eight independent chains keep the multiplier busy

28 instructions
for 8 MACs

RAW distance 8
no MUL stall

constant offsets
less address math

17.1 → 13.43 ms
per head-GEMM call

-21.5%
bit-exact

